Frequency Response of Amplifier

Low-Frequency Response (BJT) Amplifier

Low Frequency response of CE amplifier and all other capacitively coupled amplifier is limited by DC blocking and bypass capacitor.

$$X_C = \frac{1}{2\pi f C} \cong 0\ \Omega$$

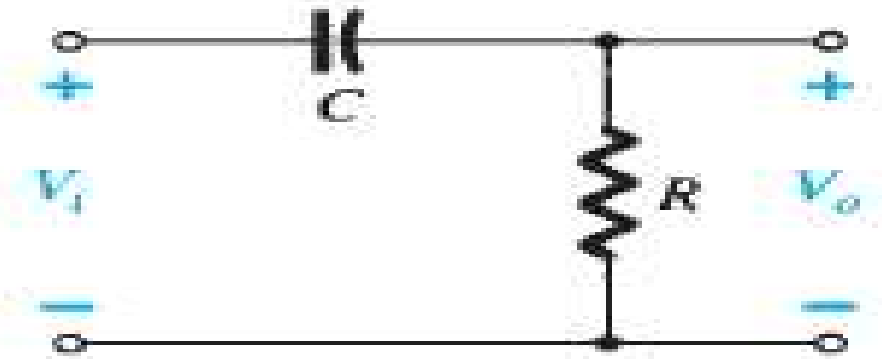
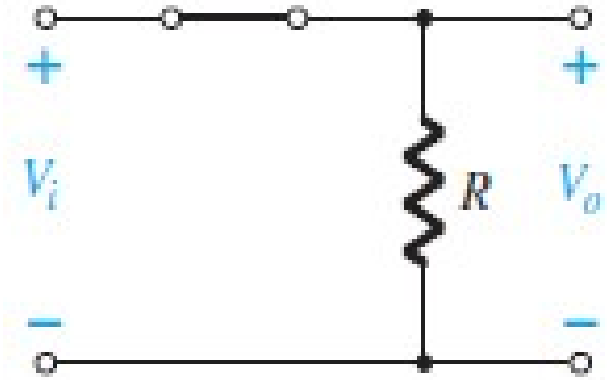
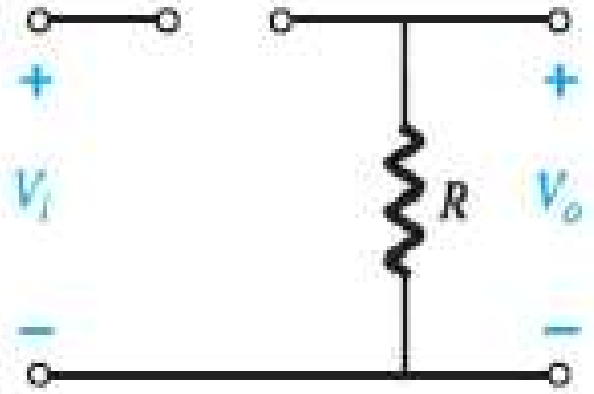


FIG. 9.14
RC combination that will define a low-cutoff frequency.

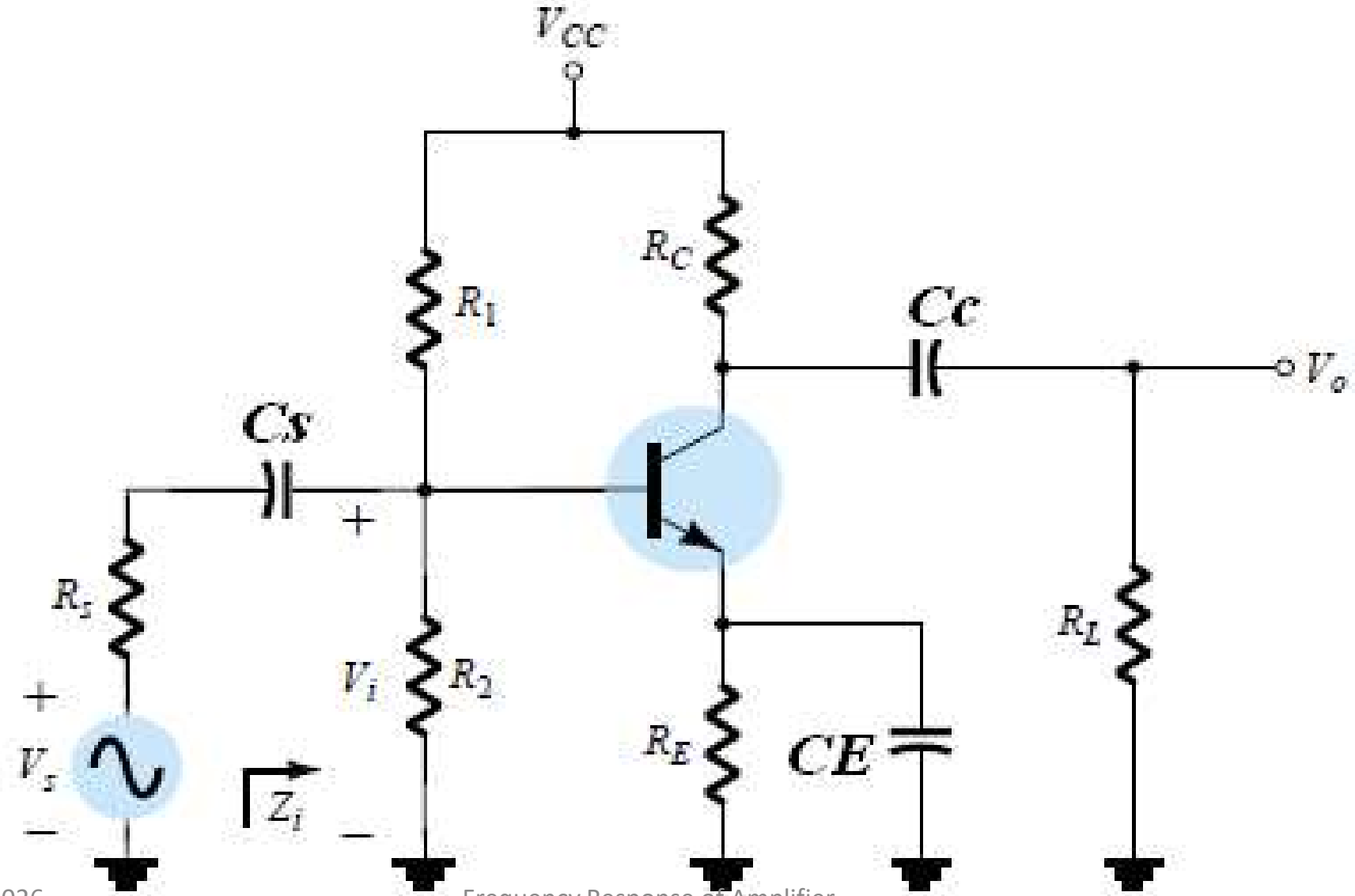
$$X_C = \frac{1}{2\pi fC} \cong 0\ \Omega$$



$$X_C = \frac{1}{2\pi fC} = \frac{1}{2\pi(0)C} = \infty\ \Omega$$



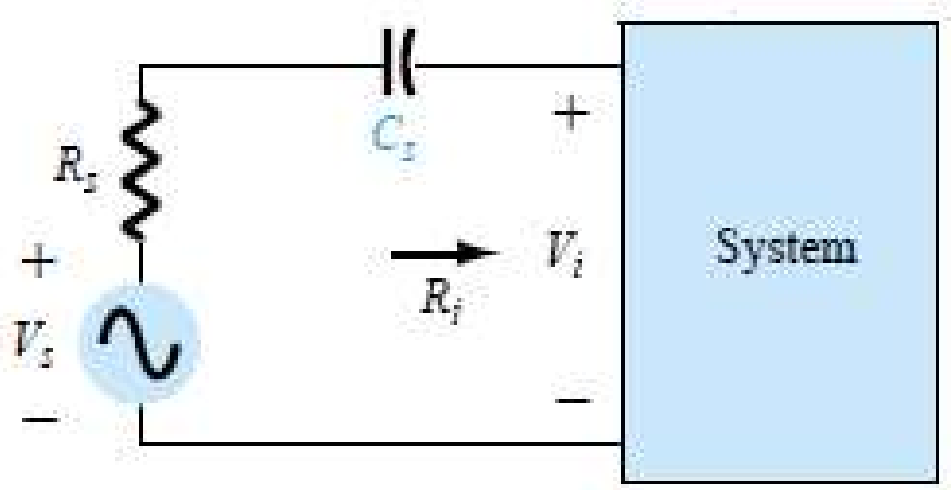
Low-Frequency Response (BJT) Amplifier



Effect of Cs

- the general form of the R-C configuration is established by the network of the following Figure. The total resistance is Rs + Ri.
- The Cutoff frequency:

$$Fls = \frac{1}{2\pi R_s}$$

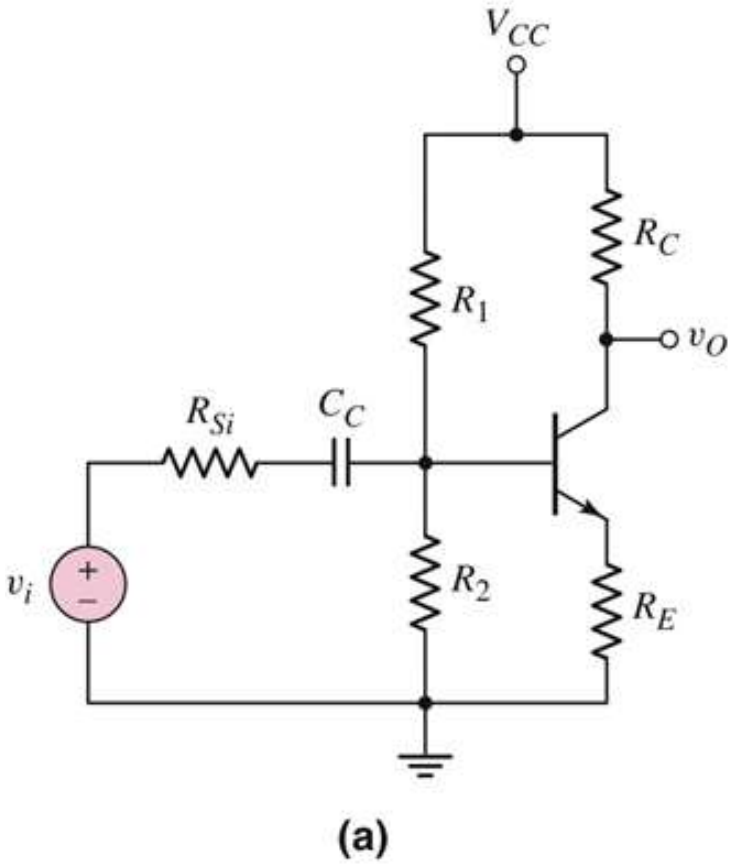


$R = R_i + R_s$

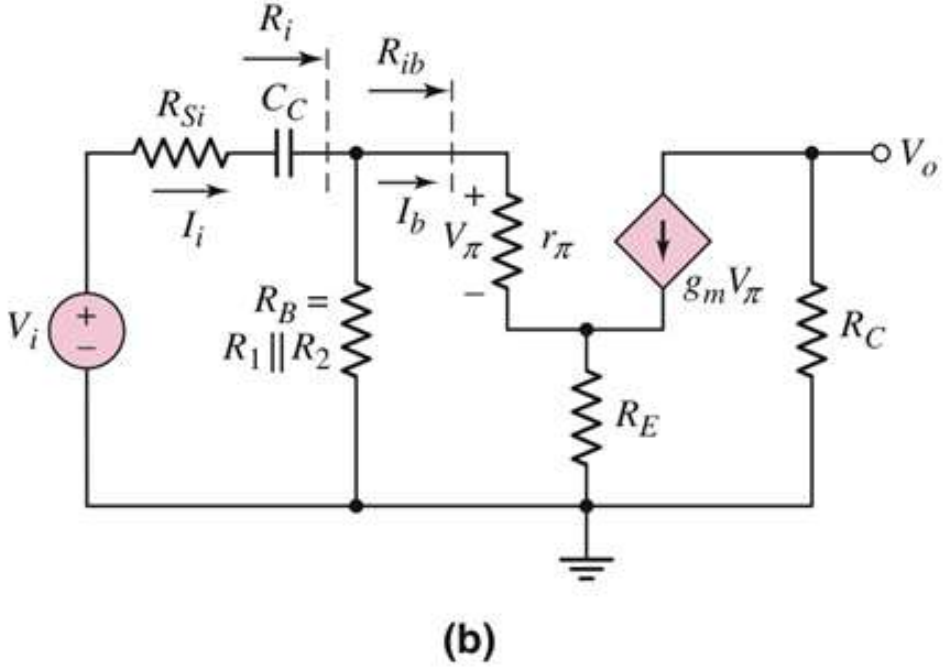
$Ri = R_{th} // r_\pi$

Common Emitter with Coupling

$$f_L = \frac{1}{2\pi(R_{Si} + R_i)C_C}$$



(a)



(b)

Small signal equivalent circuit

Common Emitter with Coupling

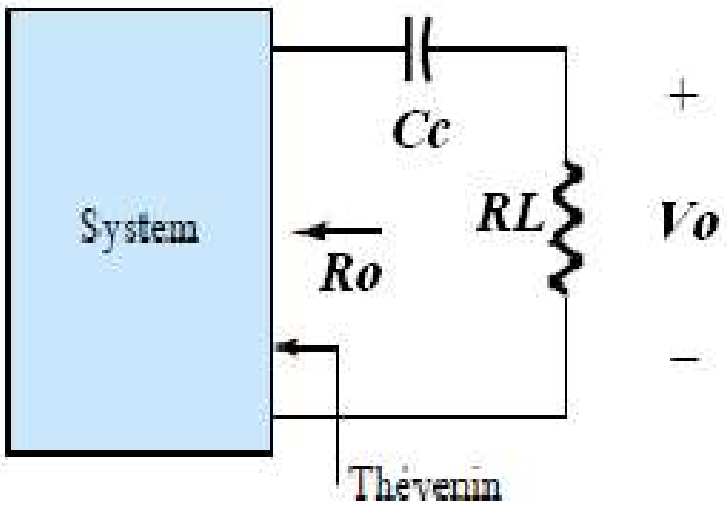
Corner frequency is $f = 1/(2\pi\tau)$ where $\tau = R_{eq}C$

R_{eq} is resistance seen by capacitor

- $R_i = R_B // [r_\pi + (1+\beta) R_E]$
- $= R_B // R_{ib}$
- Time constant for the circuit is function of equivalent resistance seen by capacitor.
- $\tau_s = (R_{si} + R_i) C_c$
- $f_L = 1/(2 \pi \tau_s)$

Effect of C_C

Since the coupling capacitor is normally connected between the O/P of the active device and the applied load, the R-C configuration that determines the low cut-off frequency due to C_C appears in the Figure.



The total series resistance is now $R_O + R_L$ and the cutoff frequency due to C_C is determined by:

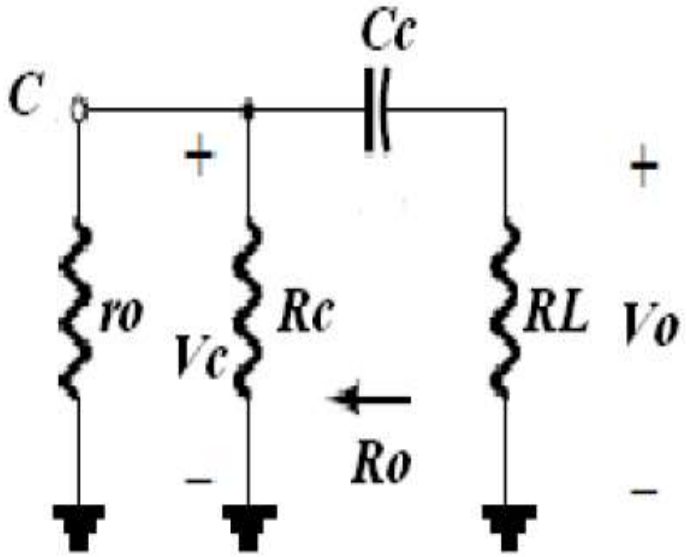
$$f_{Lc} = \frac{1}{2\pi(R_o + R_L)C_C}$$

Effect of Output Coupling Capacitor

For the network of the loaded BJT amplifier, the ac equivalent network for the O/P section with $V_i = 0$ V appears in the following Figure.

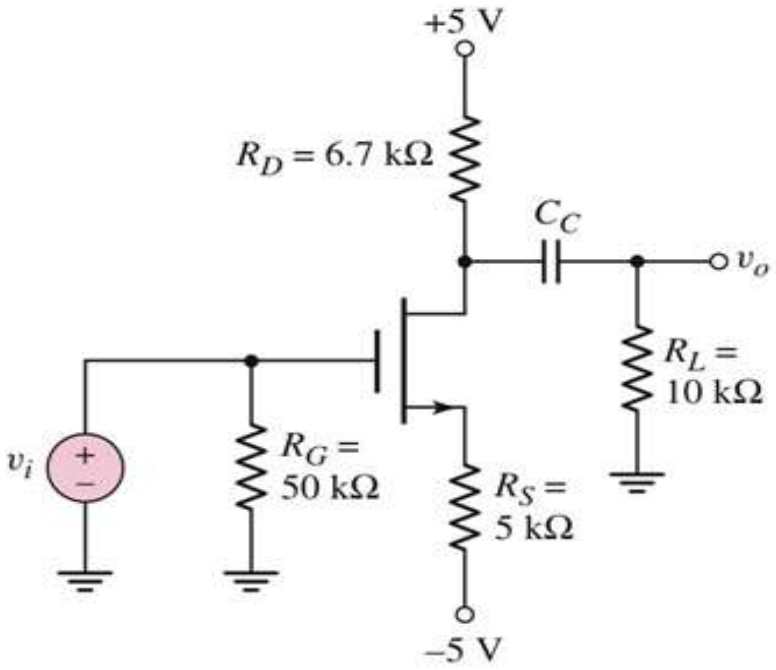
The resulting value of R**o** in the equation of f_{LC} is then simply

$$R_o = R_C // r_o$$

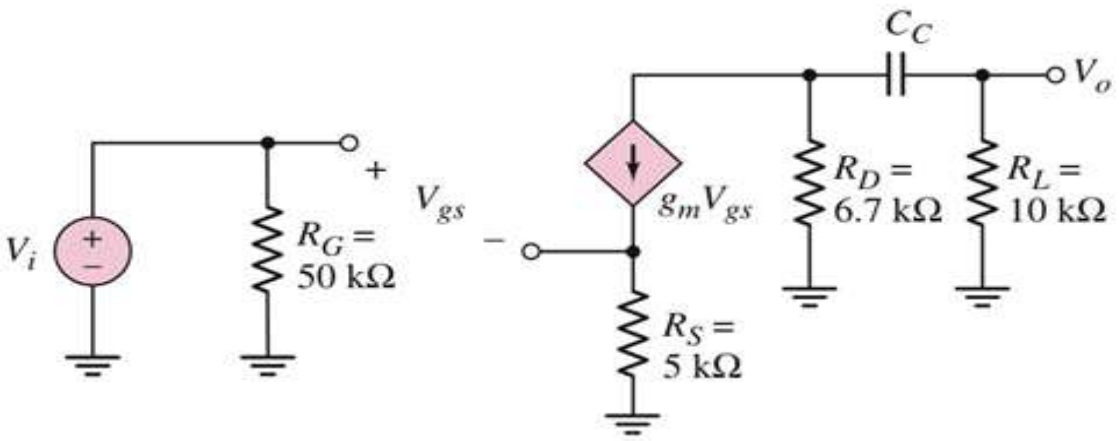


Common Source with Output Coupling Capacitor

$$f_L = \frac{1}{2\pi(R_D + R_L)C_C}$$

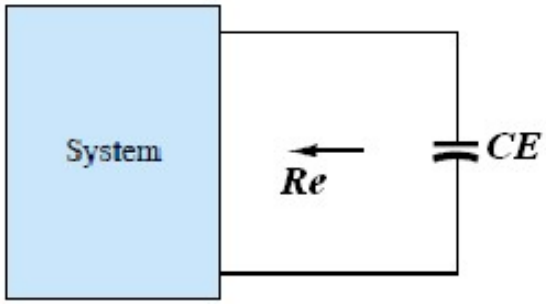


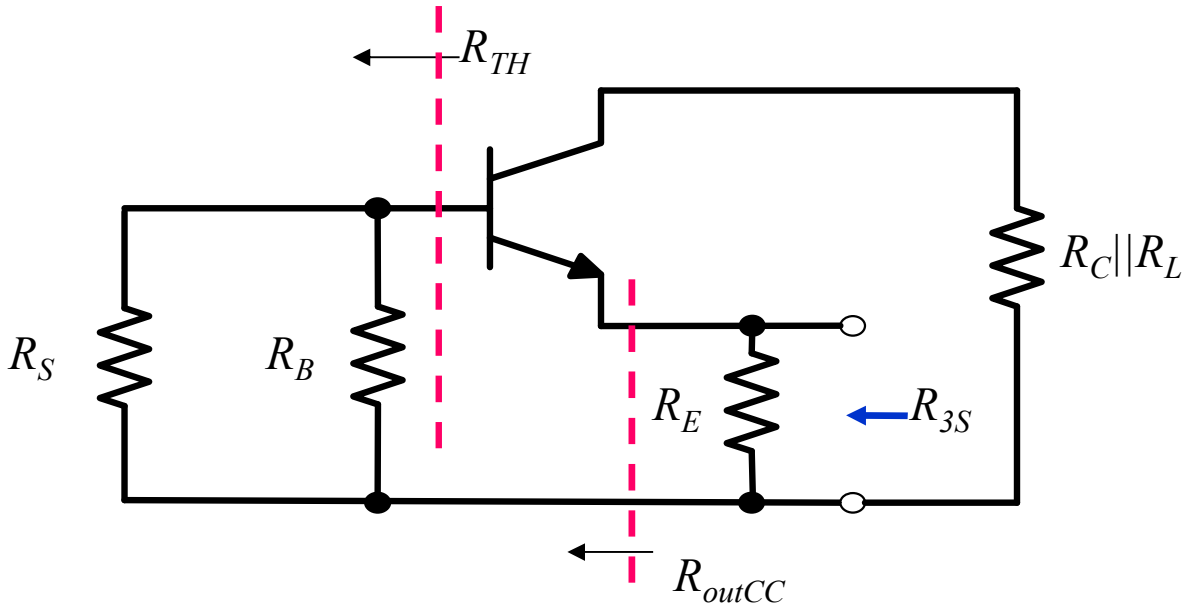
(a)



(b)

Effect of C_E





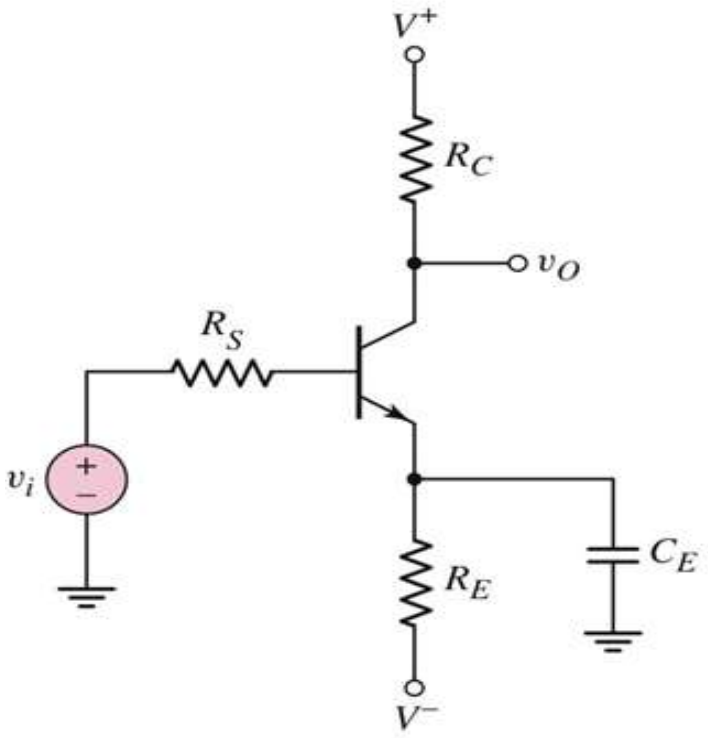
$$f_{LE} = \frac{1}{2\pi R_e C_E}$$

$$R_e = R_E \| R_{outCC} = R_E \left\| \frac{r_\pi + R_{TH}}{\beta + 1} \right.$$

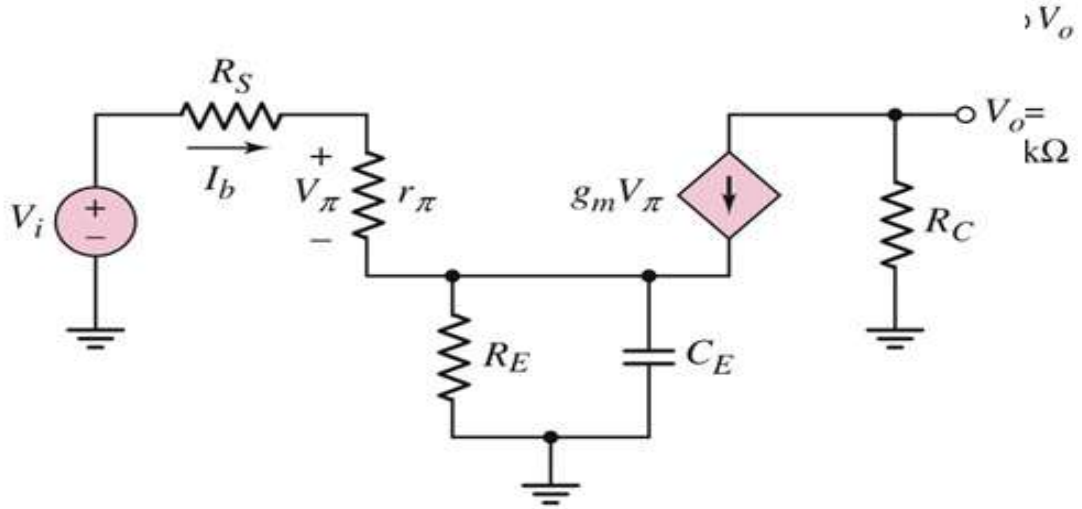
$$\frac{1}{2\pi[R_E \| \frac{[(R_S \| R_1 \| R_2) + r_\pi]}{1+\beta}] CE}$$

$$R_{TH} = R_S \| R_B$$

Emitter Bypass Capacitor



(a)



(b)

Common-emitter Amplifier

Given :

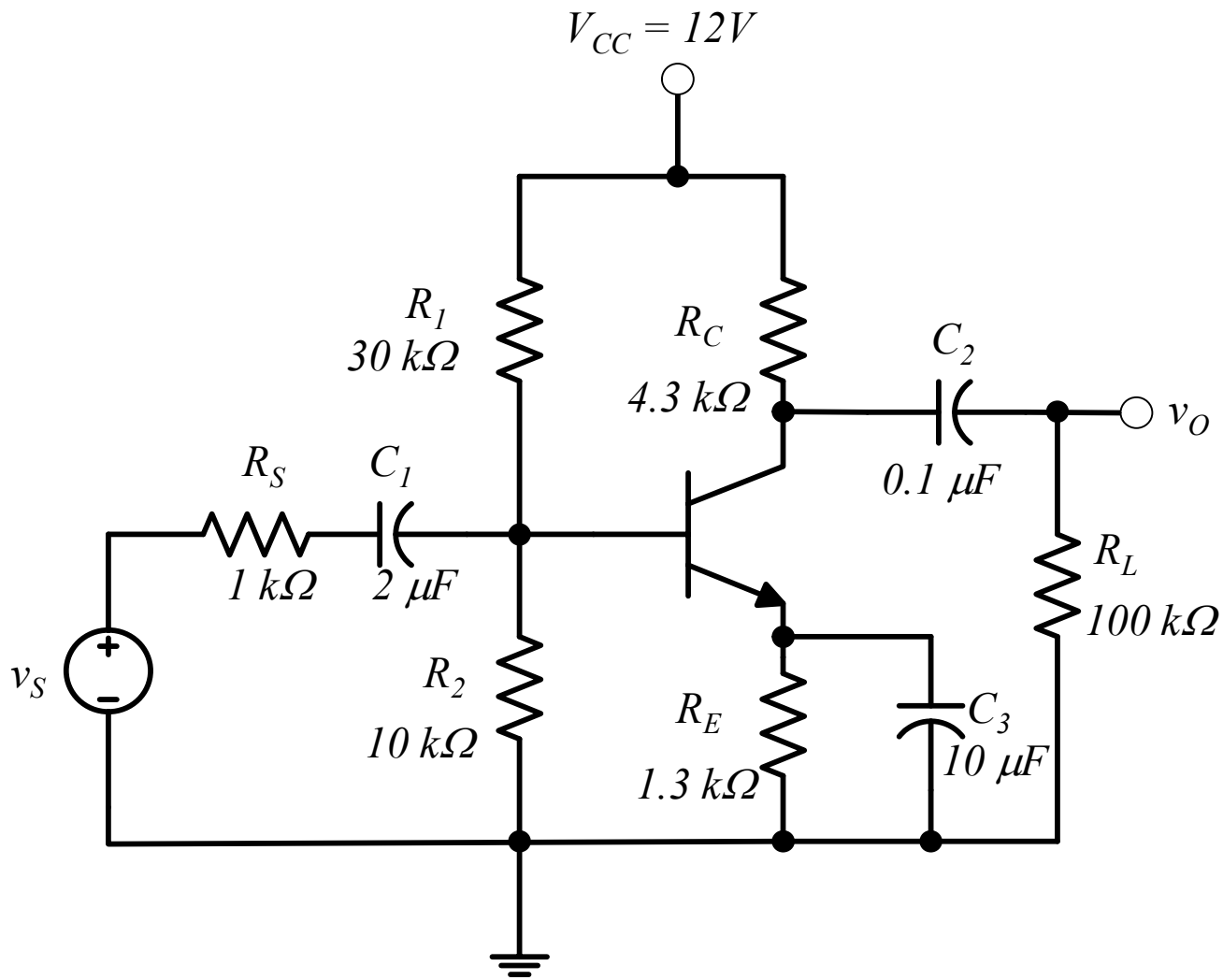
Q-point values :
1.73 mA, 2.32 V

β = 100, V_A = 75 V

Therefore,

r_π = 1.45 kΩ,

r_o =44.7 kΩ



D.C. Analysis

$$R_{TH} = R_1 \parallel R_2$$

$$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) V_{CC}$$

$$I_{BQ} = \frac{V_{TH} - V_{BE}(\text{on})}{R_{TH} + (1+\beta) R_E}$$

$$I_{CQ} = \beta \; I_{BQ}$$

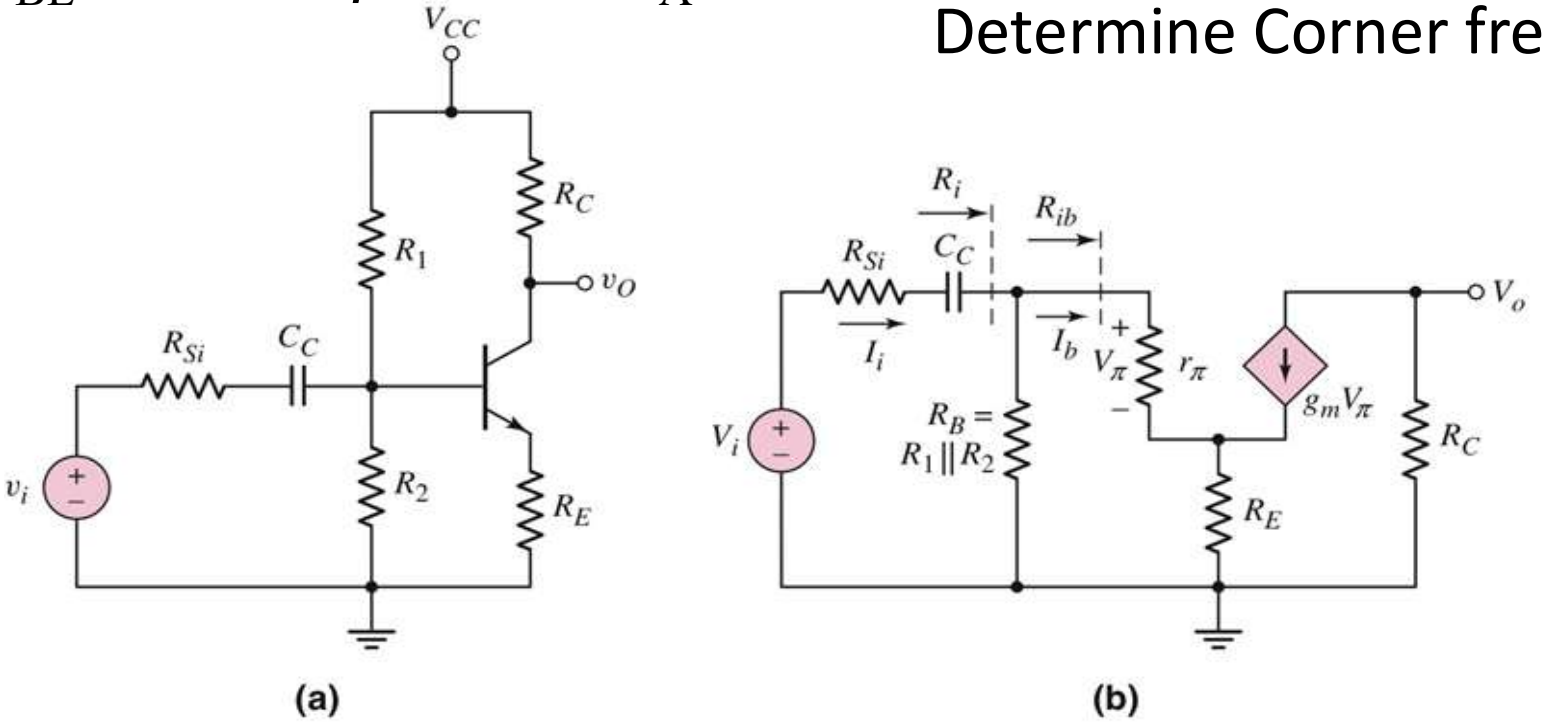
$$r_\pi = \frac{\beta V_T}{I_{CQ}}$$

$$g_m = \frac{I_{CQ}}{V_T}$$

Ex 7.3 Electronic Circuits-Neamen

R_{si} = 0.1 kΩ, $R_1 = 20$ KΩ, $R_2 = 2.2$ KΩ, $R_E = 0.1$ KΩ, $R_C = 2$ KΩ,
$C_c = 47\mu F$, $V_{CC} = 10$ V
$V_{BE} = 0.7$ V, $\beta = 200$, $V_A = \infty$

Determine Corner frequency.



$R_{TH} = R_1 \| R_2$ $R_{TH} = R_1//R_2 = 2.2// 20 = 1.98\ K\Omega$

$V_{TH} = \left(\frac{R_2}{R_1 + R_2} \right) V_{cc}$ $V_{TH} = R_2/(R_1+R_2)\ Vcc = (2.2/(2.2+20)) * 10 = 0.991\ V$

$I_{BQ} = \frac{V_{TH} - V_{BE}(\text{on})}{R_{TH} + (1+\beta) R_E}$ $I_{BQ} =$ $= (0.991 - 0.7)/ (1.98K+((1+200) *0.1)$

$= 0.0132\ mA$

$I_{CQ} = \beta\ I_{BQ}$ $I_{CQ} = \beta\ I_{BQ} = 200 * 0.0132 = 2.64\ mA$

$r_\pi = \beta\ V_T / I_{CQ} = 200 * 0.026 / 2.64 = 1.97\ K\Omega$

$r_\pi = \frac{\beta V_T}{I_{CQ}}$ $g_m = I_{CQ} / V_T = 2.64 / 0.026 = 101.4\ mA/V$

$g_m = \frac{I_{CQ}}{V_T}$ $r_{ib} = r_\pi + ((1 + \beta) R_E = 1.97 + (201 * 0.1) = 22.1K\ \Omega$

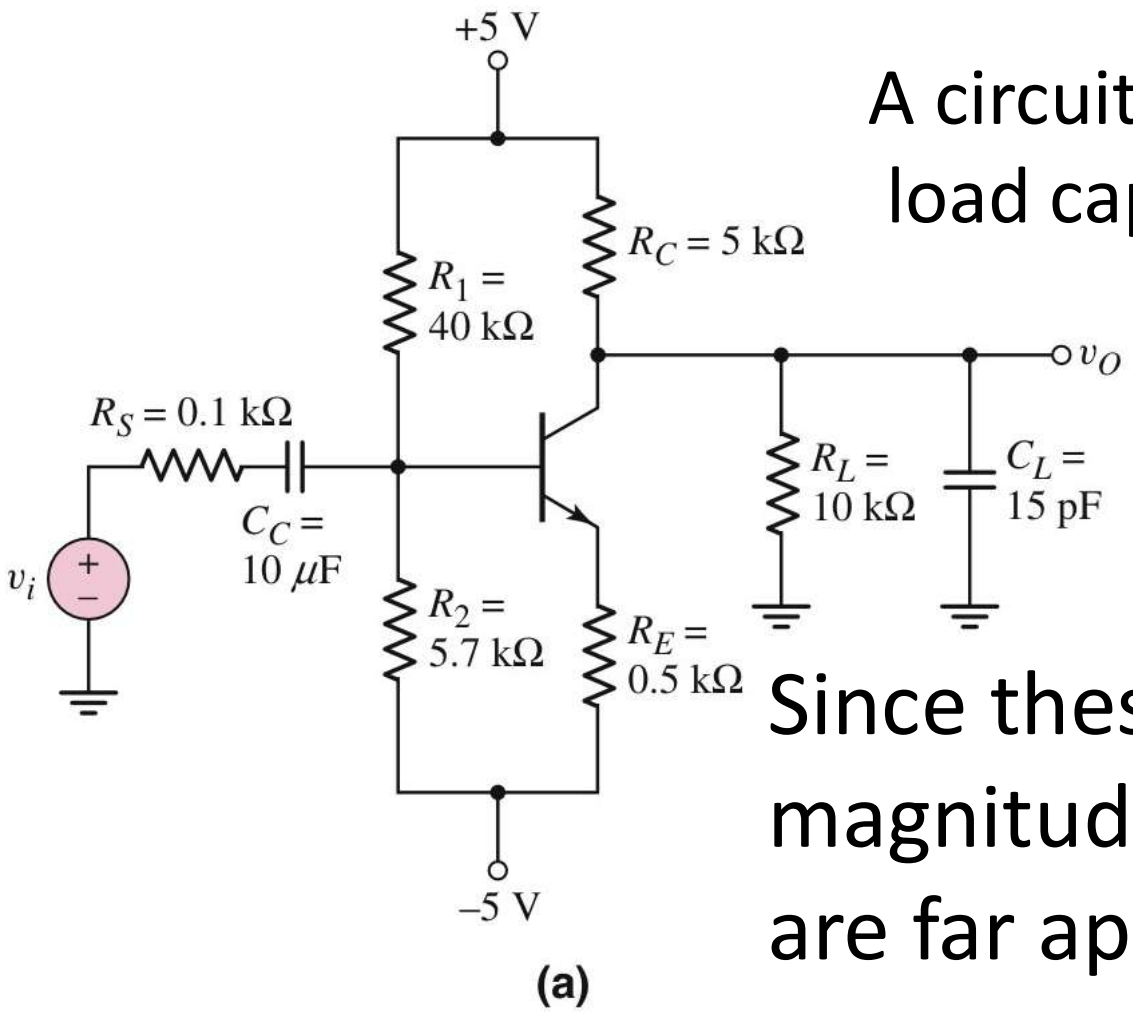
$R_{TH} = R_1//R_2 = 1.98\ K\Omega$

$r_i = R_{TH} // r_{ib} = 1.82\ K\Omega$

$$f_L = \frac{1}{2\pi(R_{Si} + R_i)C_C} = 1.77Hz$$

Example 7.7

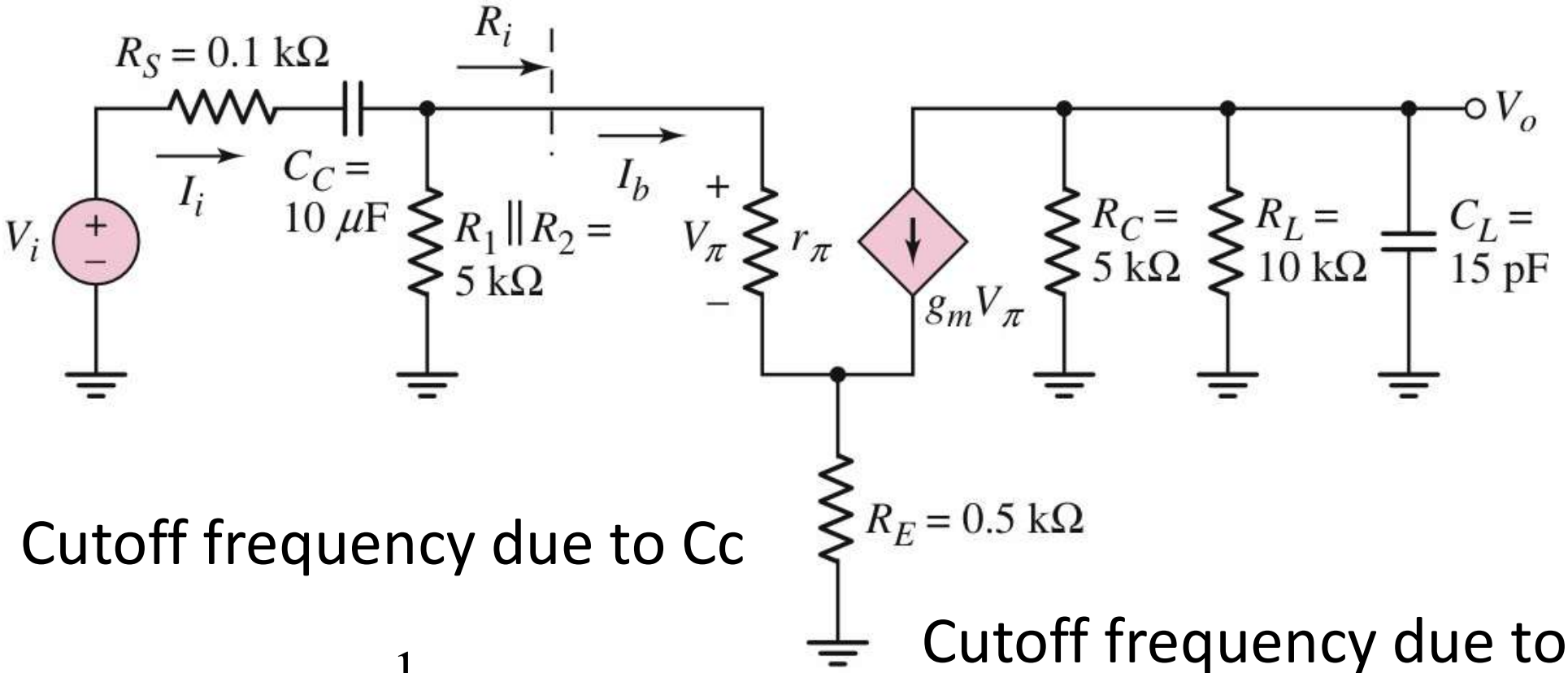
Determine mid-band gain, corner frequencies, and bandwidth of a circuit. **V_{BE}= 0.7 V, β = 100 and V_A = ∞**



A circuit with coupling capacitor and load capacitor.

Since these values differ a large in magnitude, Corner frequencies are far apart.

Example



(b)

Cutoff frequency due to Cc

$$f_L = \frac{1}{2\pi[R_S + (R_1 \| R_2 \| R_i)]C_C}$$

$$Ri = r_\pi + (1+\beta)R_E$$

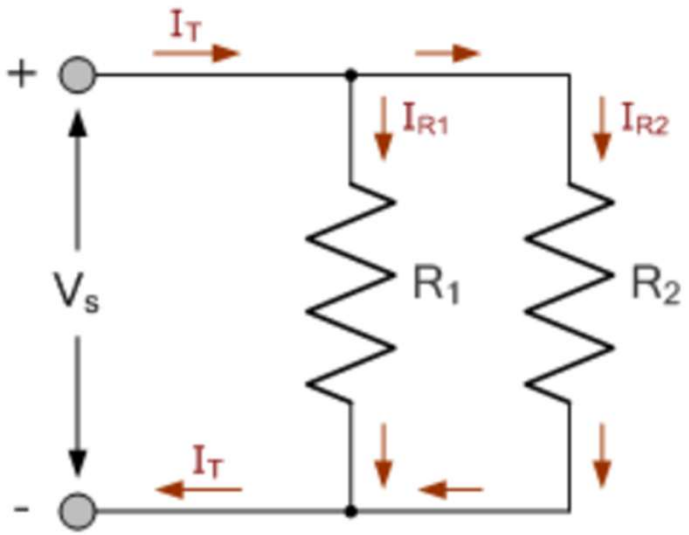
Cutoff frequency due to C_L

$$f_H = \frac{1}{2\pi(R_C \| R_L)C_L}$$

Example

- DC Analysis
- I_{CQ}= 0.99 mA
- $g_m = I_{CQ}/ V_T$ = 0.99/0.026 = 38.08 mA/V
- $r_\pi = \beta V_T / I_{CQ}$ = 100 * 0.026 / 0.99 = 2.626 KΩ
- $Rib = r_\pi + (1+\beta)R_E$ = 2.626 +(101)0.5 = 53.1 KΩ

Current Divider Rule



$$I_T = I_{R1} + I_{R2}$$

$$I_{R1} = \frac{V}{R_1} \quad \text{and} \quad I_{R2} = \frac{V}{R_2}$$

$$I_T = \frac{V}{R_1} + \frac{V}{R_2} = V\left[\frac{1}{R_1} + \frac{1}{R_2}\right]$$

$$I_{R1} = \frac{V}{R_1} = I_T\left[\frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2}}\right]$$

$$\therefore I_{R1} = I_T\left(\frac{R_2}{R_1 + R_2}\right)$$

$$\therefore V = I_T\left[\frac{1}{R_1} + \frac{1}{R_2}\right]^{-1} = I_T\left[\frac{R_1 R_2}{R_1 + R_2}\right]$$

Example -Mid-band gain

$$Av = \frac{Vo}{Vi}$$

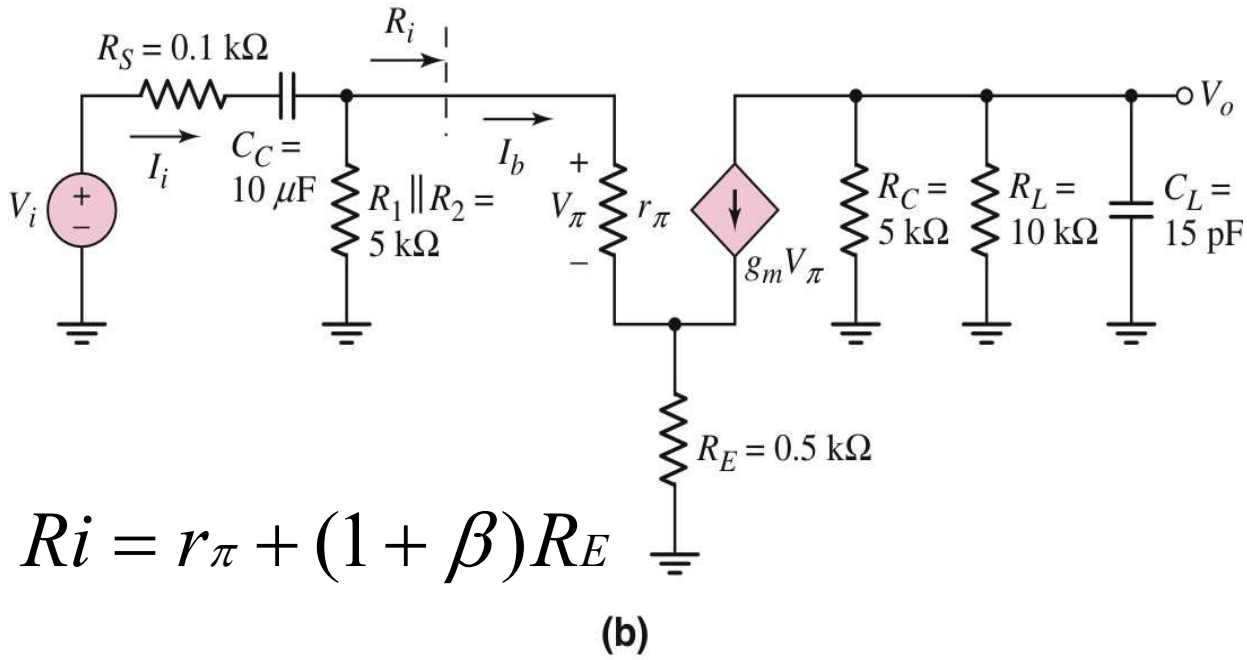
$$Vo = -g_m V\pi (RC \| RL)$$

$$V_\pi = r_\pi I_b$$

$$I_b = \left(\frac{R_1 \| R_2}{R_i + (R_1 \| R_2)}\right) Ii$$

$$I_i = \frac{Vi}{R_S + (R_1 \| R_2 \| R_i)}$$

Current Divider Rule



$$Ri = r_\pi + (1+\beta) R_E$$

$$Av = g_m r_\pi (RC \| RL)\left(\frac{R_1 \| R_2}{R_i + (R_1 \| R_2)}\right)\left(\frac{1}{[R_S + (R_1 \| R_2 \| R_i)]}\right)$$

Example_cut-off frequency

$$Av = -g_m r_\pi (R_C \| R_L)\left(\frac{R_1 \| R_2}{R_i + (R_1 \| R_2)}\right)\left(\frac{1}{[R_S + (R_1 \| R_2 \| R_i)]}\right)$$

$$= -6.14$$

$$f_L = \frac{1}{2\pi[R_S + (R_1 \| R_2 \| R_i)]C_C}$$

$\mathbf{F_L = 3.42\ Hz}$

$$Ri = r_\pi + (1+\beta)R_E$$

$$f_H = \frac{1}{2\pi(R_C \| R_L)C_L}$$

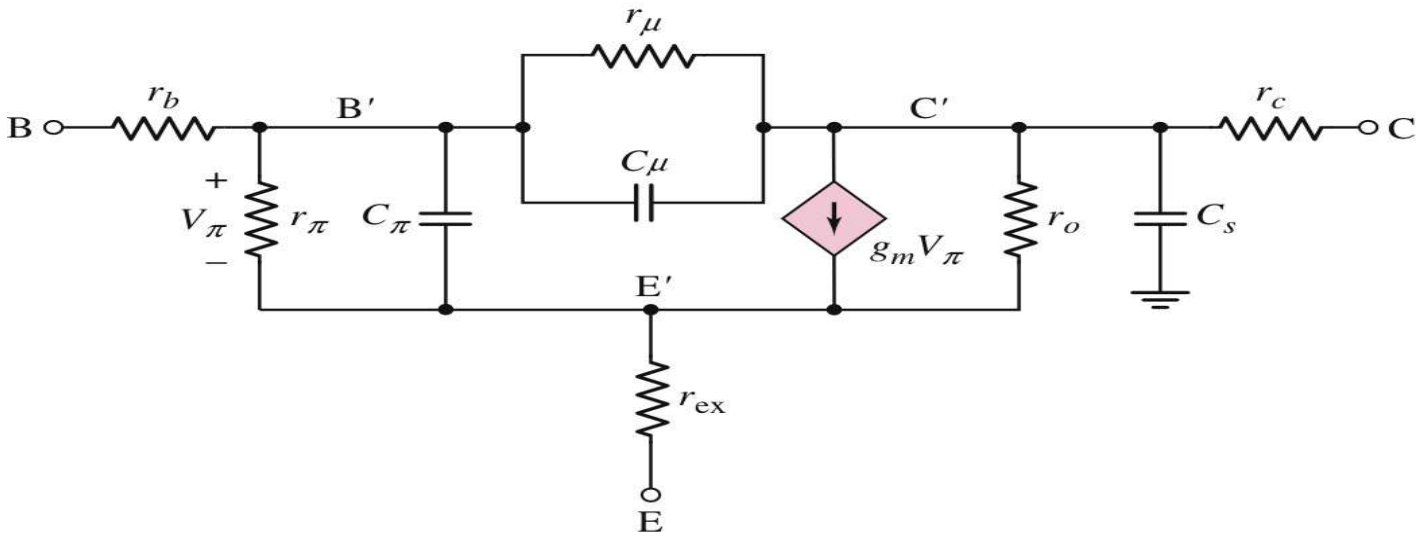
$\mathbf{F_H = 3.18\ MHz}$

Bandwidth = 3.18 MHz - 3.42 Hz ≈ 3.18 MHz

Gain –Bandwidth Product

- GB = $|A_{Vmax}|$. Bandwidth
- For fixed load capacitance, above product is constant.

Expanded Hybrid π equivalent circuit



C,B E terminals are external connections to the transister and C', B', E' are idealised internal collector ,Base and emitter regions.

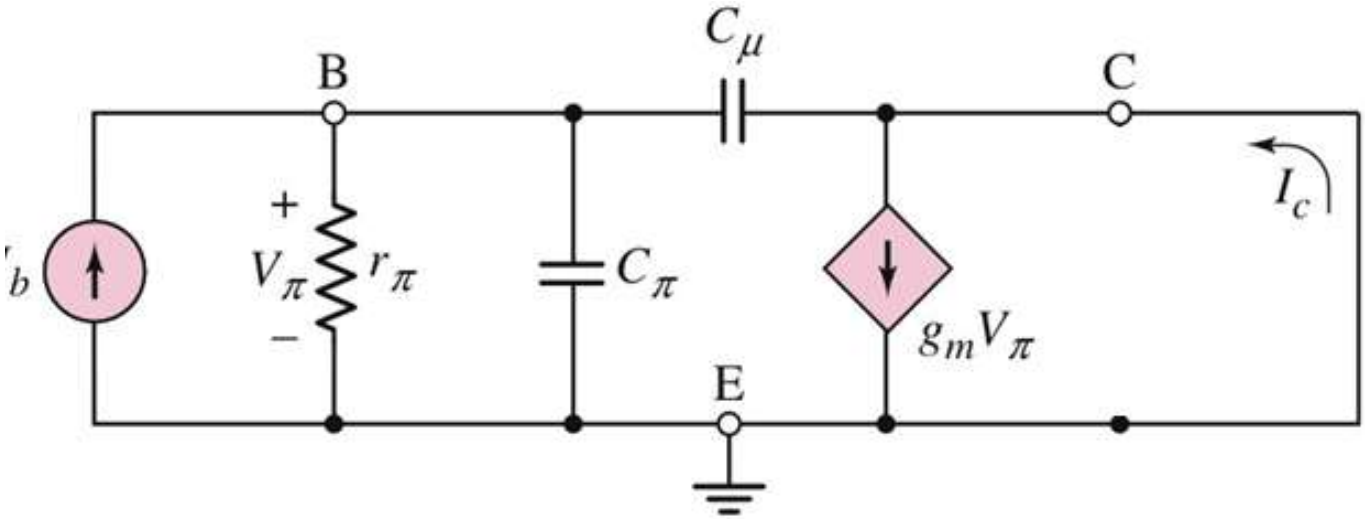
$g_m V_\pi$ is transistor collector current controlled by internal base emitter voltage.

r_o is due to early effect.

r_μ is of the order of megohm and can be neglected.

Cμ can not be neglected due to Miller Effect.

Short Circuit Current Gain



Determine Ai= Ic/Ib

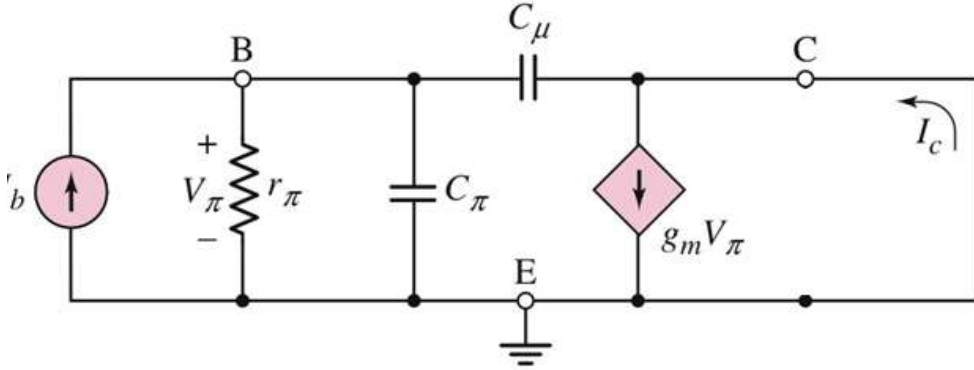
Writing a KCL equation at the input node, we find that

$$I_b = \frac{V_\pi}{r_\pi} + \frac{V_\pi}{\frac{1}{j\omega C_\pi}} + \frac{V_\pi}{\frac{1}{j\omega C_\mu}} = V_\pi\left[\frac{1}{r_\pi} + j\omega(C_\pi + C_\mu)\right] \quad \text{A}$$

Short Circuit Current Gain

From KCL equation at output node,

- $[V_\pi / (1/jwC\mu)] + Ic = g_m V_\pi$
- $Ic = V_\pi (g_m - jwC\mu)$



The input voltage V_π can then be written as

$$V_\pi = \frac{I_c}{(g_m - j\omega C_\mu)}$$

Substituting this expression for V_π into Equation (7.70) yields

$$I_b = I_c \cdot \frac{\left[\frac{1}{r_\pi} + j\omega(C_\pi + C_\mu)\right]}{(g_m - j\omega C_\mu)}$$

The small-signal current gain usually designated as h_{fe}, becomes

$$A_i = \frac{I_c}{I_b} = h_{fe} = \frac{(g_m - j\omega C_\mu)}{\left[\frac{1}{r_\pi} + j\omega(C_\pi + C_\mu)\right]}$$

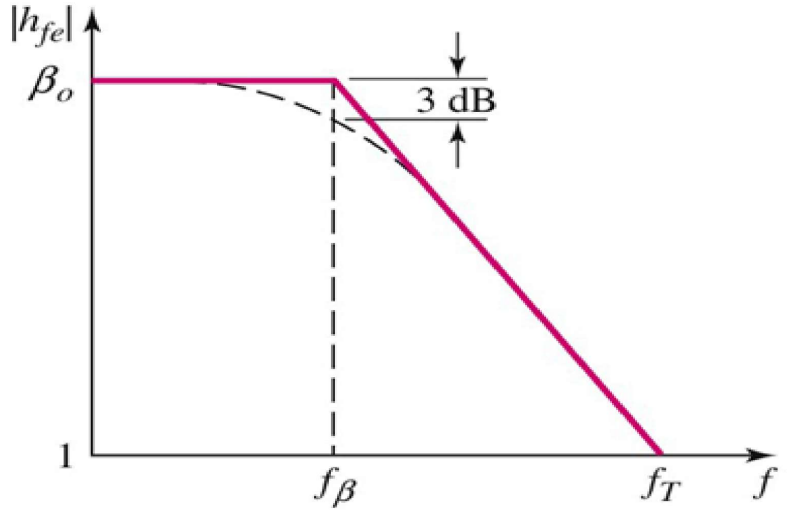
Short Circuit Current Gain

If we assume typical circuit parameter values of $C_\mu = 0.2\,\text{pF}$, $g_m = 50\,\text{mA/V}$, and a maximum frequency of $f = 100\,\text{MHz}$, then we see that $\omega C_\mu \ll g_m$. Therefore. to a good approximation, the small-signal current gain is

$$A_i = \frac{I_c}{I_b} = h_{fe} = \frac{(g_m - j\omega C_\mu)}{\left[\frac{1}{r_\pi} + j\omega(C_\pi + C_\mu)\right]}$$

$$h_{fe} \cong \frac{g_m}{\left[\frac{1}{r_\pi} + j\omega(C_\pi + C_\mu)\right]} = \frac{g_m r_\pi}{1 + j\omega r_\pi (C_\pi + C_\mu)}$$

$$f_\beta = \frac{1}{2\pi r_\pi (C_\pi + C_\mu)}$$



Magnitude of current gain decreases with increasing frequency. At F_T, which is cutoff frequency, this gain goes to 1.

Example

- Determine 3dB frequency of short circuit current gain of a bipolar transistor.
- $r_\pi = 2.6K \;\; C_\pi = 2\ pF$ and $C\mu = 0.1\ pF$

$$f_\beta = \frac{1}{2\pi r_\pi (C_\pi + C_\mu)}$$

$F_\beta = 29.1\ MHz$

- Calculate F_β and capacitance C_π of bipolar transistor. $F_T = 500$ MHz at Ic = 1mA, β_O= 100 and C$\mu = 0.3\ pF$

- $F_\beta = F_T / \beta_0 = 5\text{MHz}$
- $g_m = I_{CQ} / V_T = 38.46\text{mA/V}$
- $F_T = \frac{g_m}{2\pi(C_\pi + C_\mu)} \Rightarrow C_\pi = 11.9\ \text{pF}$

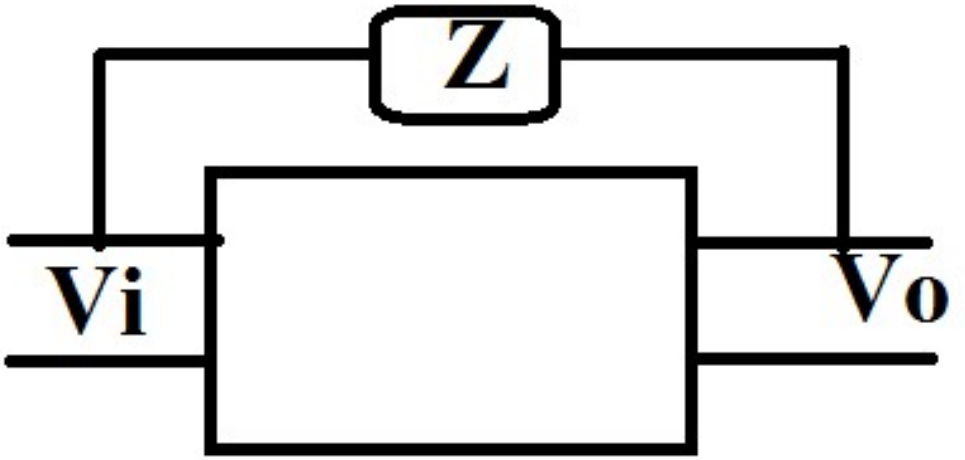


Miller Effect

- Miller capacitance was identified in 1920 in triode vacuum tubes by John Milton Miller.
- The **Miller effect** accounts for the increase in the equivalent input capacitance of an inverting voltage amplifier due to amplification of the effect of capacitance between the input and output terminals. The virtually increased input capacitance due to the Miller effect is given by $C_M=C(1+A_V)$ where A_V is the gain of the amplifier and C is the feedback capacitance.
- Although the term *Miller effect* normally refers to capacitance, any impedance connected between the input and another node exhibiting gain can modify the amplifier input impedance via this effect.

Miller Effect

- In a high-frequency Bipolar Junction Transistor (BJT) hybrid-pi model, the feedback capacitance C_μ (collector-base junction capacitance) is multiplied by the inverting voltage gain of the amplifier. This multiplication, known as the Miller effect, dramatically increases the effective input capacitance, severely limiting the high-frequency bandwidth.
- Total Input Capacitance: $C_{in} = C_\pi + C_\mu (1 + |A_v|)$

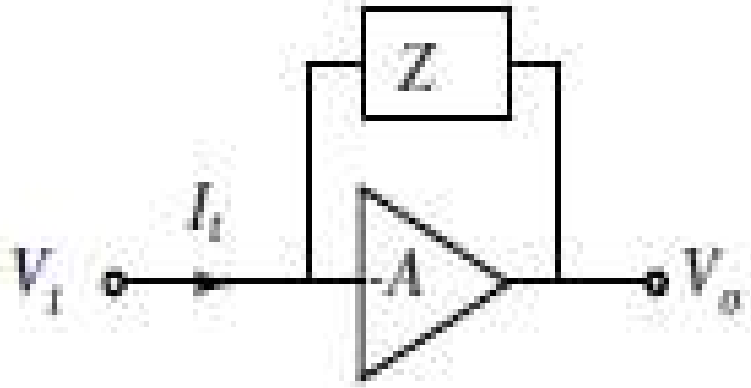


$V_0 = K V_i$



$Z1 = Z/(1 - K)$ $Z2 = Z/\ 1\text{-}\ (1/K))$

Consider the following amplifier with voltage gain $-A$, with an impedance Z connected from input to output.



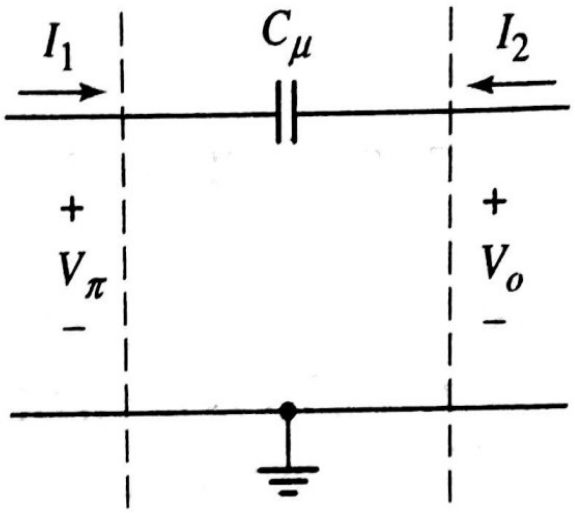
Calculating the input current

$$I_i = \frac{V_i - V_o}{Z} = V_i\left(\frac{1+A}{Z}\right)$$

The Thevenin input impedance is

$$Z_{in} = \frac{V_i}{I_i} = \frac{Z}{1+A}$$

Thus a resistor or inductor connected from input to output will look a factor of $(1+A)$ times smaller (as seen from the input terminal), and a capacitor will look $(1+A)$ times larger.

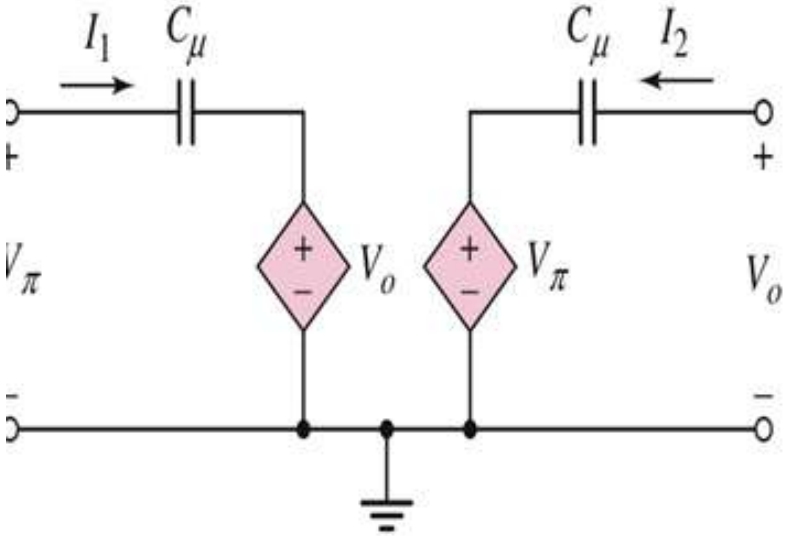


Writing KVL equations at input and output terminals,

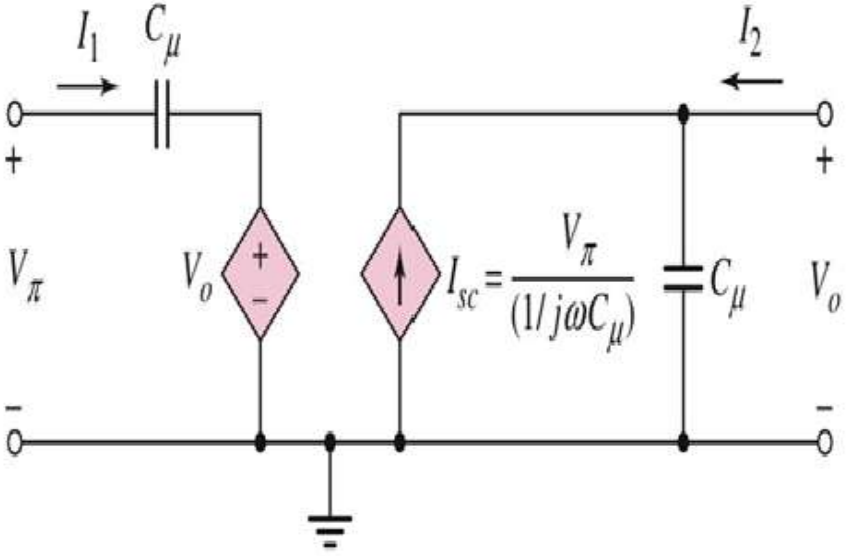
$V_\pi = I_1 (1/(j\omega C_\mu) + V_O$ ----------A

$V_O = I_2 (1/(j\omega C_\mu) + V_\pi$ ----------B

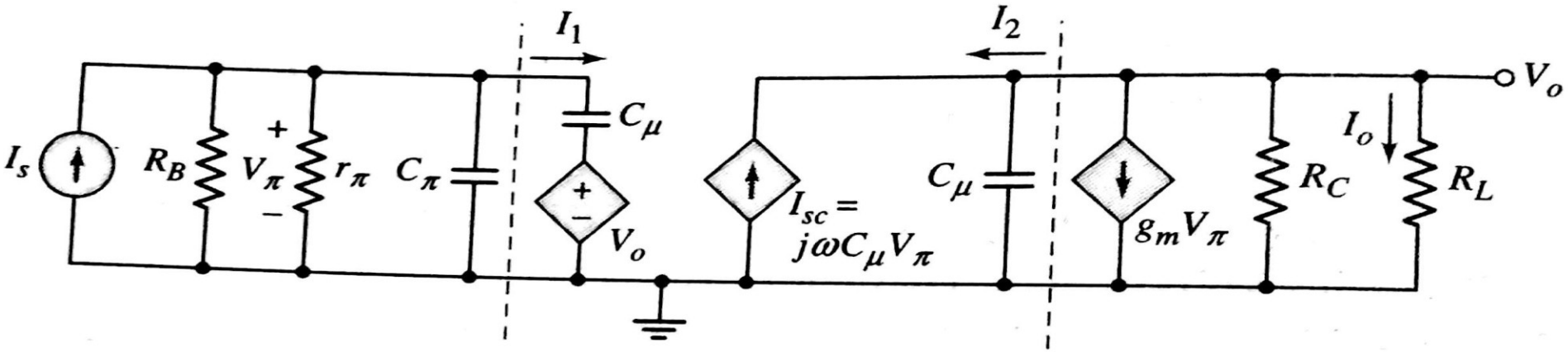
Using A & B form two port network. Convert Thevenin equivalent circuit into Norton equivalent circuit.



(a)



(b)



- Typical value of C_μ =0.2 pF g_m = 50 mA/V
- Calculate frequency at which magnitude of two dependent current sources are equal.

If $\omega C_\mu V_\pi = g_m V_\pi$, then

$f = g_m/(2\pi C_\mu) = 39.8 \times 10^9$ Hz

Since frequency of operation of bipolar transistors is far less than 40 GHz, Current source

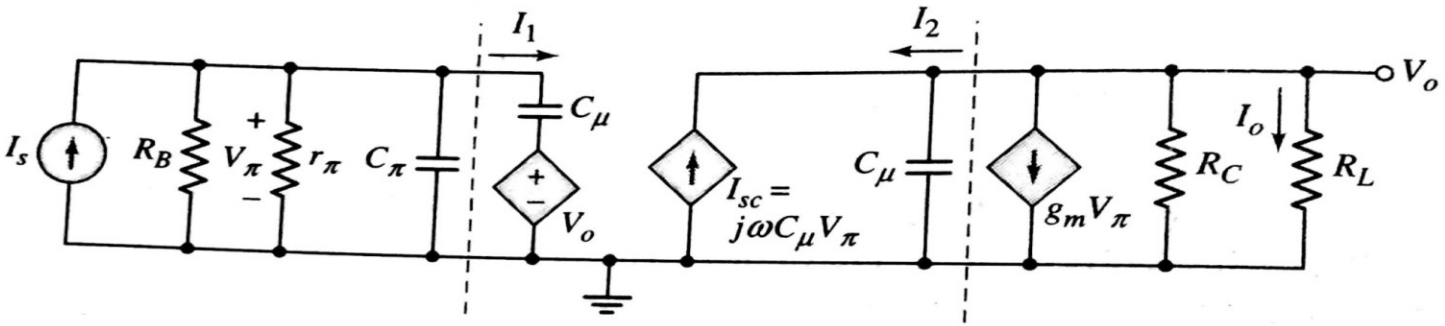
Isc = $j\omega C_\mu V_\pi$ is **negligible** compared to $g_m V_\pi$ source.

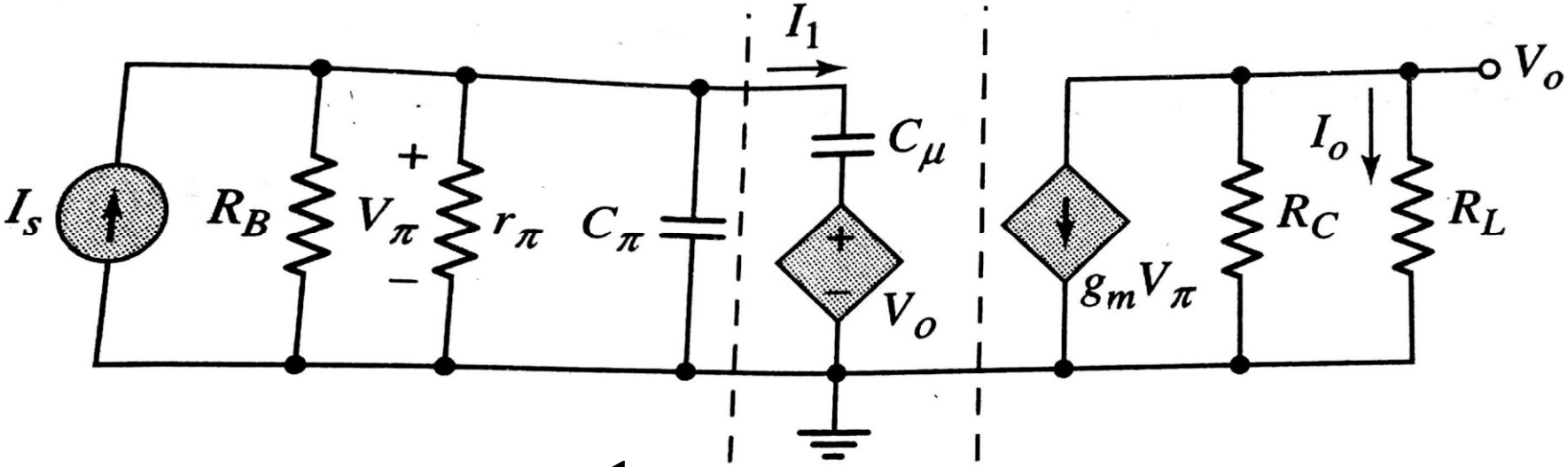
Approximations

- Calculate frequency at which magnitude of impedance C_μ is equal to $R_C \| R_L$
- If $R_C \| R_L = \frac{1}{\omega C_\mu}$, then $f = \frac{1}{2\pi C_\mu (R_C \| RL)}$
- If we assume $R_C = R_L = 4\ K\Omega$, $C_\mu = 0.2\ pF$

$$f = \frac{1}{2\pi C_\mu (R_C \| RL)} = \frac{1}{2\pi (0.2 * 10^{-12})[(4 * 10^{-3} \| 4 * 10^{-3})]} = 398\ MHz$$

- If frequency of operation is much smaller than 400 MHz then C_μ will be much greater than $(R_C \| R_L)$ and C_μ can be considered to be open circuit.
- Using approximations, circuit reduces to



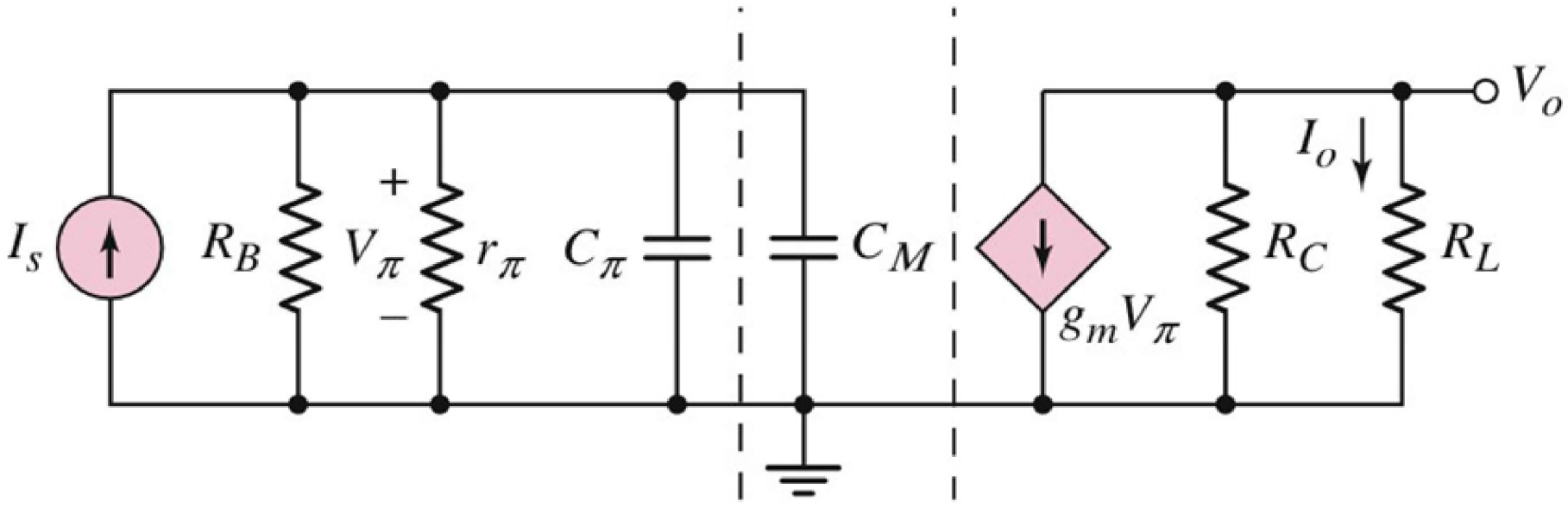


$$I_1 = (V_\pi - V_O) / \left(\frac{1}{j\,\omega C_\mu}\right) = j\,\omega C_\mu\,(V_\pi - V_O)$$

$$V_O = -\,g_m\,V_\pi\,(R_C \| RL)$$

$$I_1 = j\,\omega C_\mu\,[\,1 + g_m(R_C \| RL)]\,V_\pi$$

Small signal equivalent circuit, including Miller capacitance

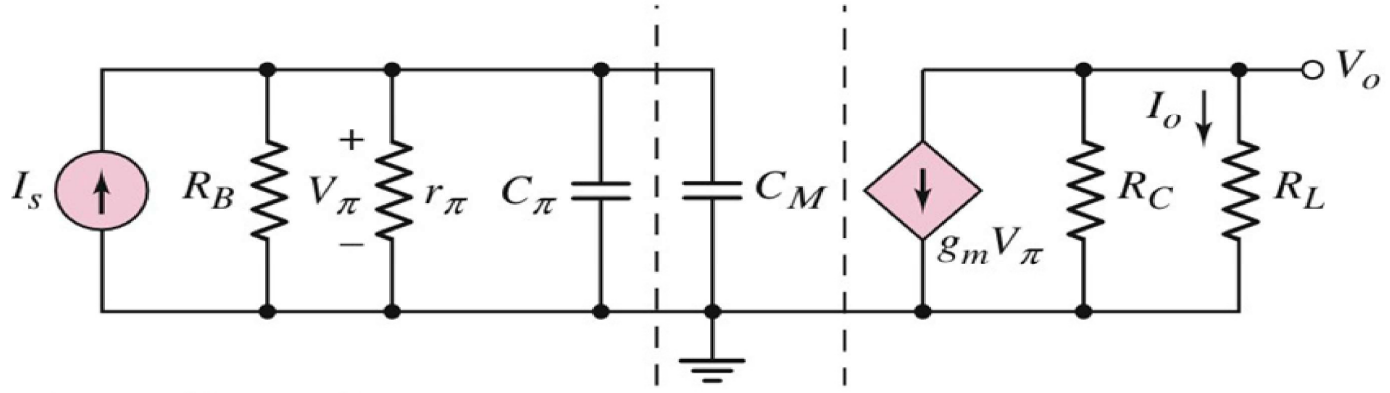


$$C_M = C_\mu \ [\ 1 + g_m(R_C \| RL)]$$

- C_M is called as Miller capacitance and multiplication effect of C_μ is the Miller effect.

Example

- Determine 3 dB frequency of the current gain, both with and without the effect of C_M



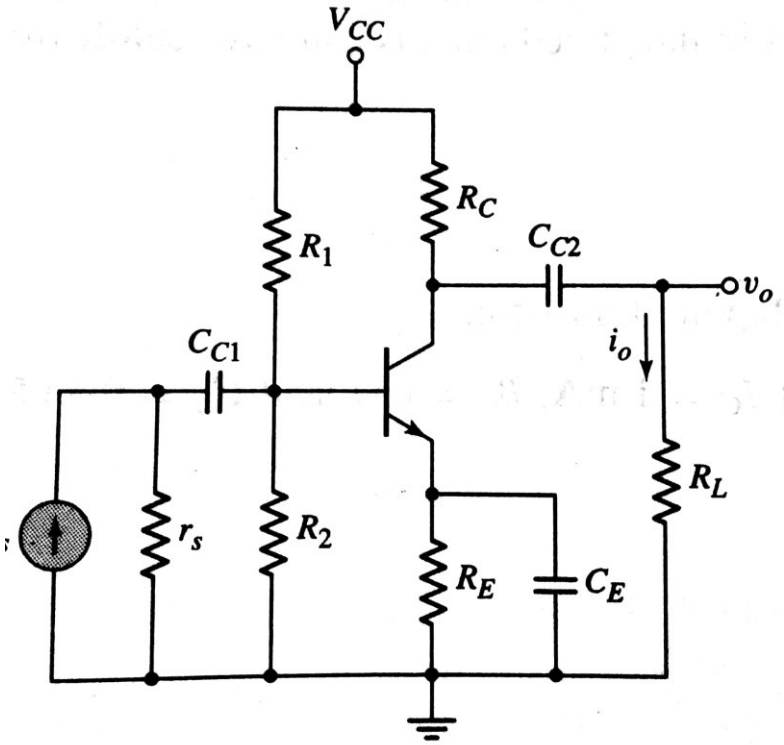
- $R_C = R_L = 4K\Omega$ $r_\pi = 2.6\ K\Omega$ $R_B = 200\ K\Omega$ $C_\pi = 4pF$
  $C_\mu = 0.2pF$ $g_m = 38.5\ mA/V$

Example

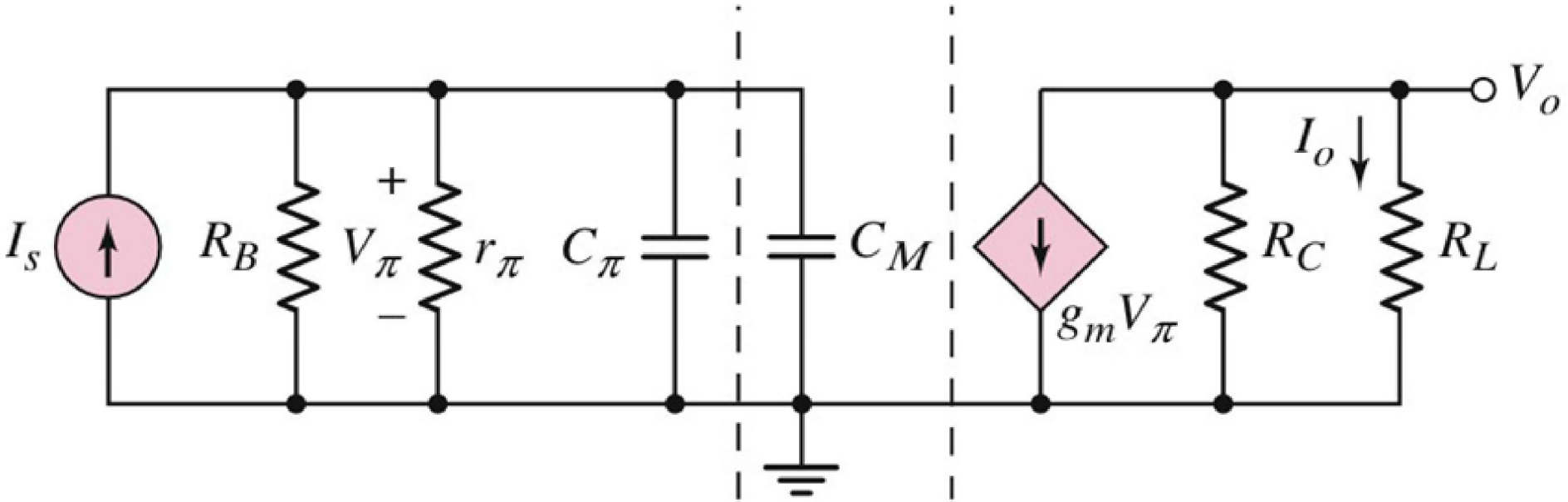
- $A_i = I_O/I_S$
- $I_O = - g_m V_\pi (R_C /(R_C + R_L))$
- $V_\pi = I_S [R_B \| r_\pi \| \frac{1}{j\omega C_\pi} \| \frac{1}{j\omega C_M}] = I_S [\frac{R_B \| r_\pi}{1 + j\omega (R_B \| r_\pi)(C_M + C_\pi)}]$
- $A_i = I_O/I_S = - g_m (R_C /(RC + R_L))[\frac{R_B \| r_\pi}{1 + j\omega (R_B \| r_\pi)(CM + C_\pi)}]$
- 3 dB frequency is $[\frac{1}{2\pi (R_B \| r_\pi)(CM + C_\pi)}]$
- Neglecting effect of C_M
- $f_{3dB} = [\frac{1}{2\pi (200 * 10^3 \| 2.6 * 10^3)(4 * 10^{-12})}] =$ **15.5MHz**
- Miller capacitance $C_M = C_\mu [1 + g_m (R_C \| RL)] = 15.6$ pF
- $f_{3dB} = [\frac{1}{2\pi (R_B \| r_\pi)(CM + C_\pi)}] =$ **3.16 MHz**

.

Ex 7.11



R_1=200KΩ R_2= 220KΩ R_C=2.2KΩ
R_L= 4.7KΩ r_S = 100 KΩ V_{CC}= 5 V
The transistor parameters are---
V_{BE} = 0.7 V, β = 100, V_A =∞ C_π =10pF
C_μ = 2pF using simplified hybrid-π model calculate Miller capacitance, and 3 dB frequency



$R_B = R_S \parallel R_1 \parallel R_2$

$C_M = C_\mu \ [\ 1 + g_m(R_C \parallel RL)] = 109\ pF$

3 dB frequency is $\left[\frac{1}{2\pi(R_B \parallel r_\pi)(CM+C_\pi)}\right] = 0.506MHz$

TYU 7.6

- I_{CQ}, β_0 , F_β , C_μ are given. Calculate C_π.

- $F_\beta = \frac{1}{2\pi r_\pi (C_\pi + C_\mu)}$
- $g_m = I_{CQ} / V_T$
- $F_\beta = F_T / \beta_0$
- $F_T = \frac{g_m}{2\pi(C_\pi + C_\mu)}$

TYU 7.8

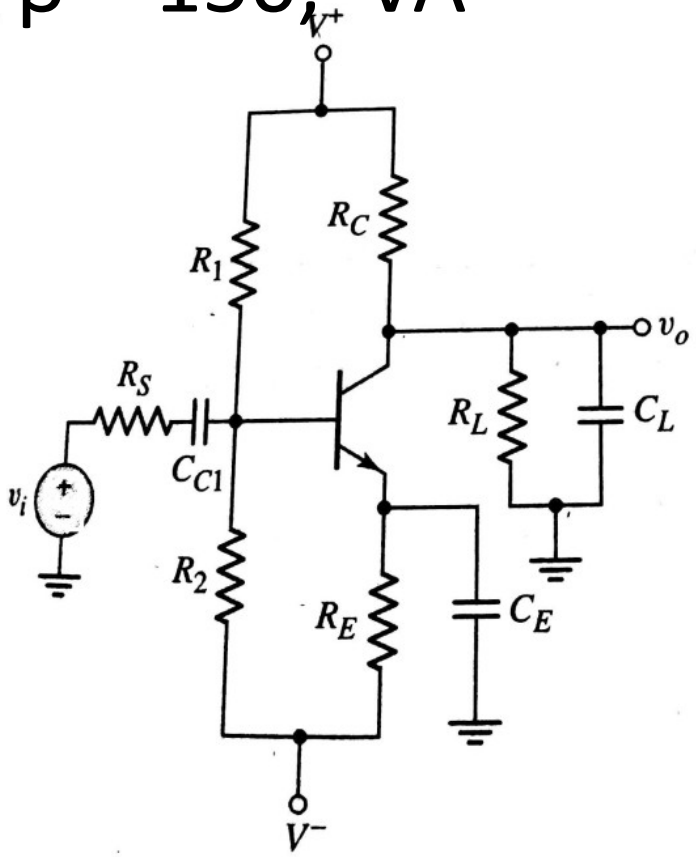
- r_π ,β_0 , F_T , C_μ are given. Calculate C_π & F_β.

$F_\beta = F_T / \beta_0$

- $F_\beta = \frac{1}{2\pi r_\pi (C_\pi + C_\mu)}$

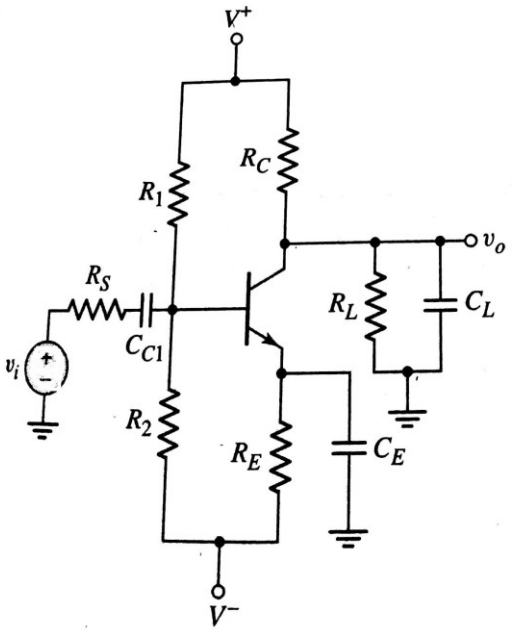
Example 7.14 Determine upper corner frequency and midband gain of CE circuit. R1=40KΩ, R2= 5.72KΩ, RC=5 KΩ, RE=0.5 KΩ, RL= 10 KΩ, RS = 0.1 KΩ, V+CC= 5 V, V−CC= 5 V The transistor parameters are--- VBE = 0.7 V, β = 150, VA =∞

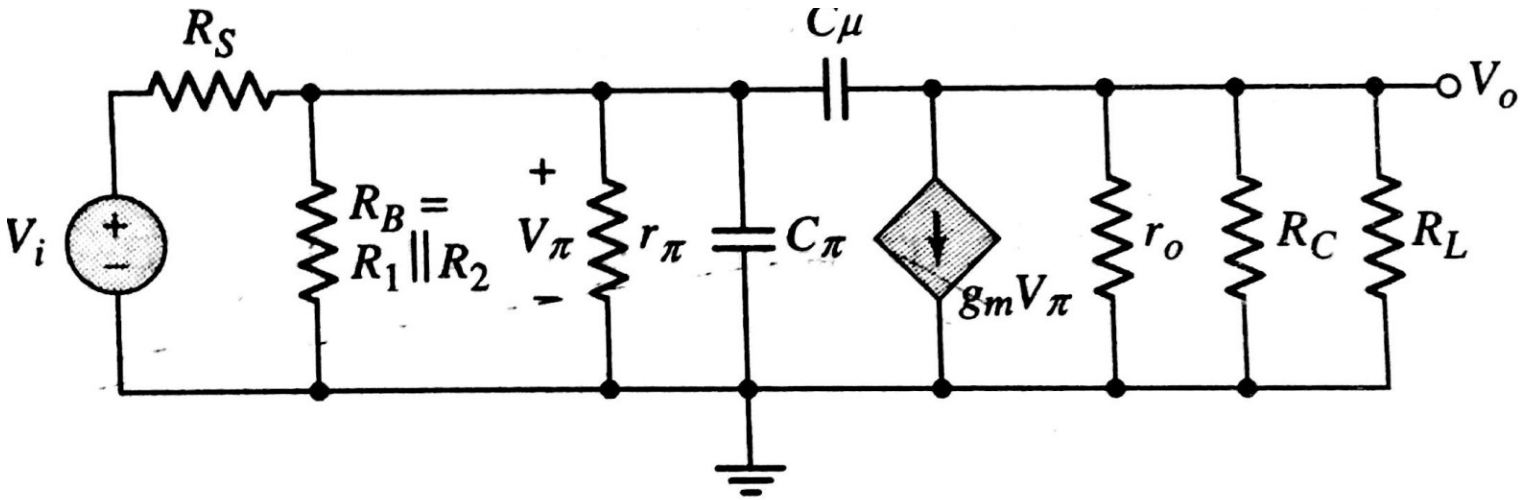
Cπ =35 pF Cμ = 4pF

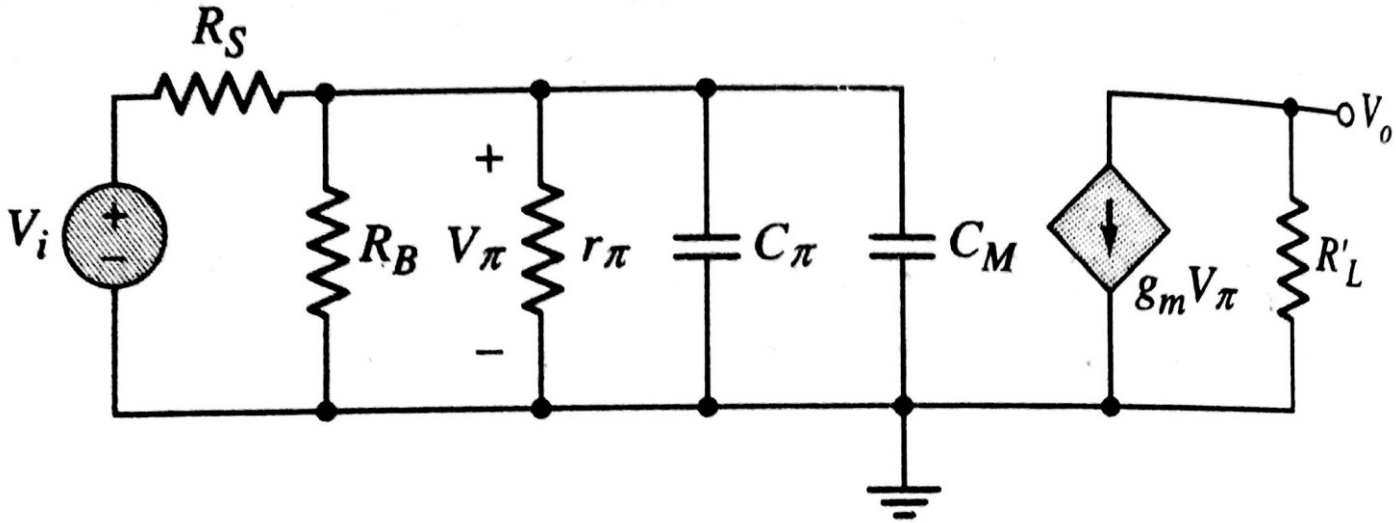


Example 7.14 Determine upper corner frequency and midband gain of CE circuit.
R1=40KΩ, R2= 5.72KΩ, RC=5 KΩ, RE=0.5 KΩ, RL= 10 KΩ, RS = 0.1 KΩ, V+CC= 5 V, V−CC= 5 V

The transistor parameters are--- VBE = 0.7 V, β = 150, VA =∞ Cπ =35 pF Cμ = 4pF



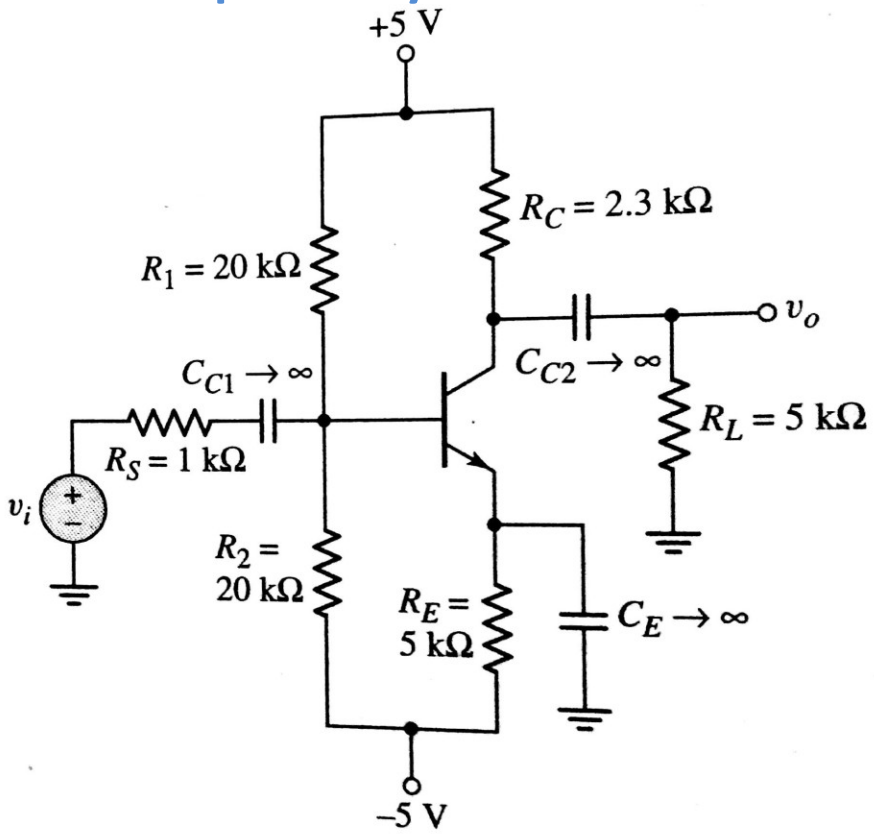




Example 7.14

- I_{CQ}= 1.03 mA
- $g_m = I_{CQ}/ V_T$ = 1.03/0.026 = 39.6 mA/V
- $r_\pi = \beta V_T / I_{CQ}$ = 150 * 0.026 / 1.03 = 3.79 KΩ
- $C_M = C_\mu [1 + g_m(R_C \| RL)]$
- $$f_H = \frac{1}{2\pi[r_\pi \| R_B \| R_S](C_\pi + C_M)} = 2.94 MHz$$
- Midband gain = $g_m R'_L \left[\frac{(R_B \| r_\pi)}{(R_B \| r_\pi)+R}\right]$ = 126

Ex 7.14 $V_{BE} = 0.7$ V, $\beta = 125$, $V_A = 200$V $C_\pi = 24$ pF $C_\mu = 3$pF Calculate Miller capacitance, upper corner frequency and midband gain.



$R_{TH} = R_1 // R_2 = 10$ KΩ

$V_{TH} = 0$ V

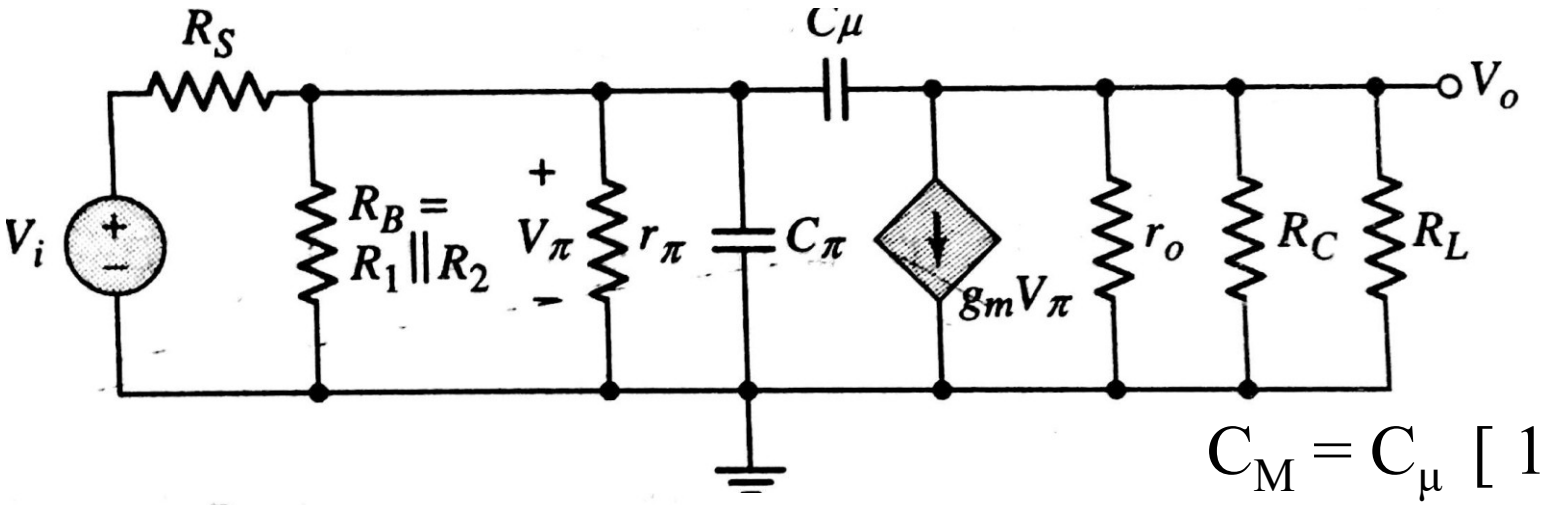
$I_{BQ} = (0 - 0.7 - (-5))/(10 + ((1+125) * 5) = 0.00672$mA

$I_{CQ} = \beta I_{BQ} = 0.84$ mA

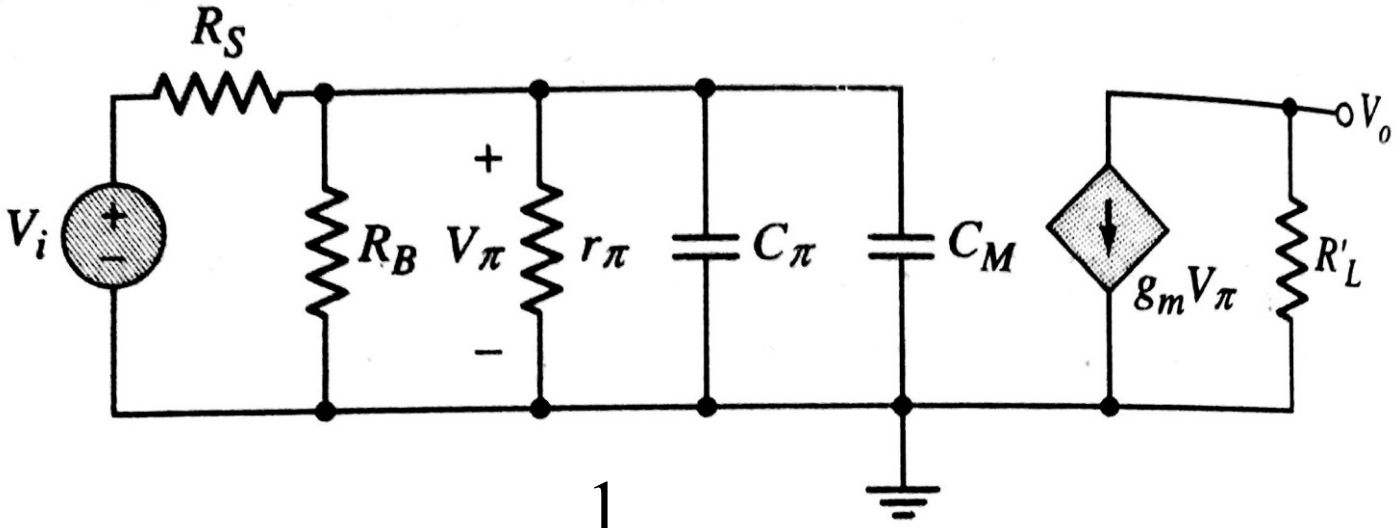
$r_\pi = \beta V_T / I_{CQ} = 125 * 0.026 / 0.84 = 3.87$ KΩ

$g_m = I_{CQ} / V_T = 0.84 / 0.026 = 32.3$ mA/ V

$r_O = V_A / I_{CQ} = 238$ KΩ



$$C_M = C_\mu \, [\, 1 + g_m (r_o \| R_C \| RL)]$$



= 155pF

$$f_H = \frac{1}{2\pi[r_\pi \| R_B \| R_S](C_\pi + C_M)}$$

= 1.21 MHz

Midband gain = $g_m R'_L$

$$\left[\frac{(R_B \| r_\pi)}{(R_B \| r_\pi) + RS}\right]$$

= -37.3

Reference

- Neaman